Mu Yuan

240-930-3436 |vincentyuan227@gmail.com | https://www.linkedin.com/in/mu-yuan227/

EDUCATION

University of Maryland, College Park College Park, Maryland

Bachelor of Science in Computer Science, Minor in General Business ***Expected Graduation: Dec 2026***

Honors: 2024 OMSE Academic Excellence Awards

Relevant Coursework: Algorithms, Advanced Data Structure, Object Oriented Programming, Computer Systems, Web Application Development with JavaScript, Machine Learning, Computer Vision, Artificial Intelligence, Software Engineering

PROFESSIONAL EXPERIENCE

Everstay Tech Shenzhen, CN

Machine Learning Intern ***May 2024 – Aug 2024***

- Built **multimodal LLM reranker** with **PyTorch** and **JAX**, improving relevance on production benchmarks and evaluation
- Applied **image and video tokenization**, compressing inputs, boosting **throughput**, and reducing **GPU memory** usage
- Implemented **instruction-tuning** and **DPO/IPO** pipelines, increasing helpfulness and **factuality** under evaluation metrics
- Optimized **decoding** with **KV cache** and **speculative decoding**, reducing **p95 latency** significantly under production workloads
- Designed experiments and statistical analyses, running **A/B tests** with **confidence intervals** and **calibration metrics** effectively
- Integrated **open-vocabulary** understanding and **safety guardrails**, strengthening **multimodal** reliability in production systems
- Collaborated cross-functionally with product and infrastructure teams, delivering **reproducible pipelines** and **evaluation suites**

Aerodynamics Laboratory Remote

Research Assistant ***Sep 2022 – Jan 2023***

- Developed **Python-based data processing pipeline** standardizing **5,000+ aerial infrared images** and **annotations**
- Enabled **reproducible ML model training** and **validation** through **automated data preprocessing** workflows
- Engineered **image augmentation algorithms** expanding **training datasets** by **3x** applying **domain-specific transformations**
- Improved **model accuracy** by **15%** on **thermal imagery classification** through **advanced data augmentation**
- Created **analytical tools** for **evaluating model performance** processing **confusion matrices** and **mAP metrics**

RELEVANT PROJECTS

NLP Educational Chatbot ***2025***

- Built a **domain-specific chatbot** to answer **NLP course concepts**, serving as an interactive teaching assistant
- Designed and implemented a **RAG pipeline** with **chunking, embeddings, and FAISS retrieval** for evidence-based responses
- Fine-tuned and applied **Transformer models** to deliver context-aware explanations of **definitions, algorithms, and formulas**
- Developed a **FastAPI backend** with **Streamlit interface**, supporting real-time queries and multi-turn conversations
- Evaluated quality using **Recall@k, F1-score, and human ratings**, achieving an average answer score of **4.5/5**
- Deployed a demo capable of handling **100+ NLP-related queries**, demonstrating reliable accuracy and usability

Record Management System ***2024***

- Developed **full-stack application** with **Node.js/Express backend** and **React/Redux frontend** for **asset management**
- Implemented **comprehensive JWT authentication** and **role-based access control** with **2s initial load time**
- Built **RESTful** and **GraphQL APIs** with **modular architecture** supporting **complex search operations**
- Integrated **AWS S3** for **secure content delivery** and **version control** with **media processing workflows**
- Containerized **application components** with **Docker** orchestrating **deployment** on **AWS ECS** with **blue-green strategy**
- Implemented **automated scaling policies** and **infrastructure monitoring** reducing **operational incidents** by **76%**

TECHNICAL SKILLS

Languages: Java, Python, C/C++, MATLAB, JavaScript, HTML/CSS, SQL, R, OCaml, Ruby, Rust, Kotlin, TypeScript

Frameworks/Libraries: React.js, Node.js, Express.js, RESTful API, Vue.js, jQuery, Angular, Flask, Django, Spring Boot, Redux

Tools: Git, Docker, Kubernetes, AWS (EC2, S3, ECS), GitLab CI/CD, Unix/Linux, Jenkins, Agile, Selenium, Terraform, Appium

Databases: PostgreSQL, MySQL, MongoDB, Redis, Database Sharding, Read Replicas

Architecture & Design: Microservices, Distributed Systems, Event-Driven Architecture, Circuit Breaker Patterns